

Astra and Opus

Five challenging parts

Prepared by Ryan | 7 October 2026

Independent block layouts and realized hexahedral meshes for five selected CAD geometries. Each model received the same geometry, layout preferences and fixed acceptance checks.

GPT 6 Astra

5 / 5 counted

\$15.07 API equivalent | 61.55 min

Opus 5.5

4 / 5 counted

\$31.97 CLI usage | 176.72 min

All ten final meshes pass geometry checks.

One Opus attempt remains rejected by the fixed command audit. Its logs indicate an apparent false positive from echoed command text; the rejection has been retained. Each model used 10 full mesh revisions in total.

HOW TO READ THIS REPORT

Each following page shows the original CAD, followed by the two model results: block layout, aligned top-down mesh and isometric mesh. SJ (scaled Jacobian) measures element quality; larger values are better, with a fixed minimum of 0.2. The shared color range is 0.2 to 1. Low SJ is yellow; high SJ is lavender. Statistics include all cells; mesh colors show exposed surfaces.

Optimised layouts (added 9 October 2026). The fourth column shows each layout after the corner optimiser (hexlayoutopt), which keeps the model's blocks and moves only their corners. It replaces the drawn layout only if every check passes and the worst element is no lower: 8 of 10 layouts improved (median min SJ +0.10); 2 are kept as drawn.

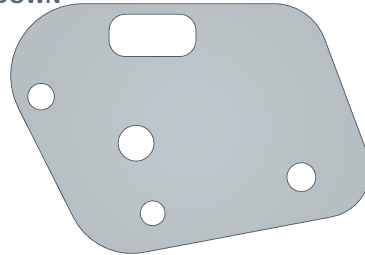
Selected showcase, not a population benchmark: these geometries had previously verified meshes, which were withheld from authors. Astra used medium reasoning effort; Opus used high. Usage figures have different accounting bases and are not subscription invoices. No human quality grades were assigned.

Rounded plate with bores and slot

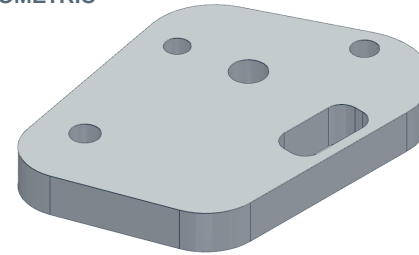
Original CAD

Unmodified input geometry.
Top and isometric views use the same orientation as the meshes below.

TOP DOWN



ISOMETRIC



Views and quality

Top-down meshes are projections of the actual 3D mesh, not separate 2D meshes.
Both densities were independently checked.

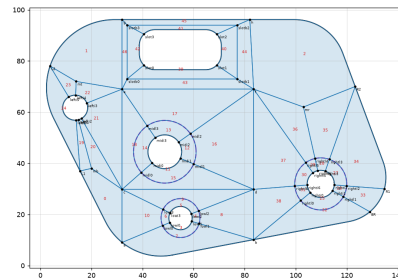
GPT 6 Astra

Counted pass

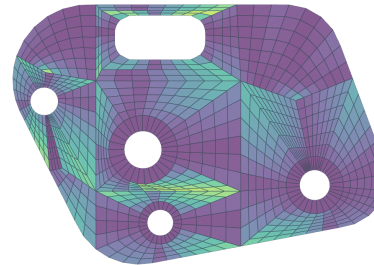
47 blocks | 5,086 hexes
Min SJ: 0.3055
Doubled SJ: 0.3055
4 full revision(s) | 17.56 min
Usage: \$4.66

Standard API equivalent

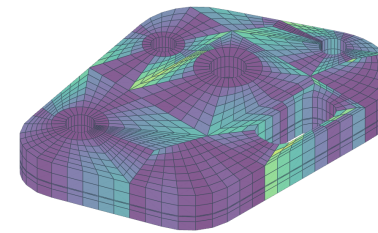
BLOCK LAYOUT



TOP-DOWN MESH

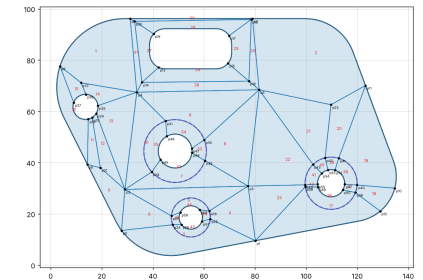


ISOMETRIC MESH



OPTIMISED LAYOUT

Min SJ 0.4431, fails 3-D aspect (27.1 > 20)
Not accepted: the drawn layout stands



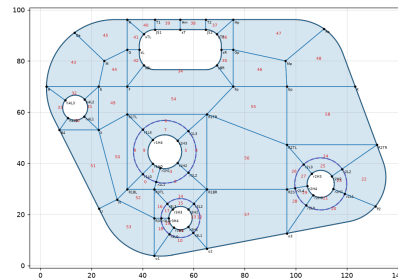
Opus 5.5

Counted pass

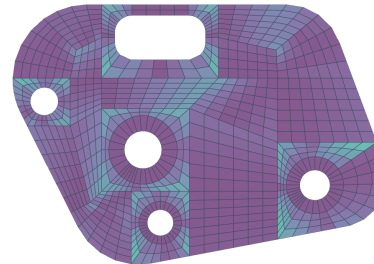
59 blocks | 4,484 hexes
Min SJ: 0.5661
Doubled SJ: 0.5661
5 full revision(s) | 52.68 min
Usage: \$9.54

Claude CLI reported usage

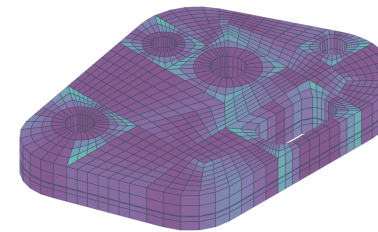
BLOCK LAYOUT



TOP-DOWN MESH

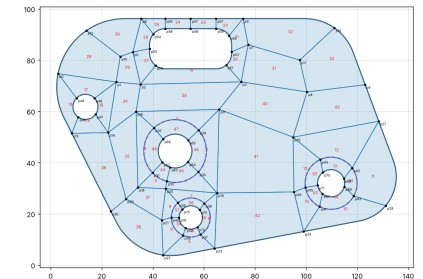


ISOMETRIC MESH



OPTIMISED LAYOUT

Min SJ 0.5661 → 0.6864 | doubled 0.6864



0.2 floor



SJ color scale (all meshes)

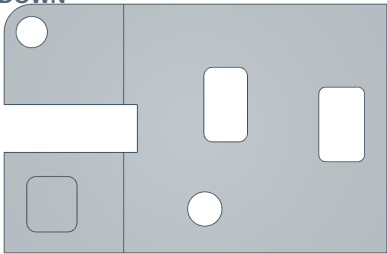
Exposed faces colored by cell SJ; interior cells included in metrics.

Notched plate with stepped features

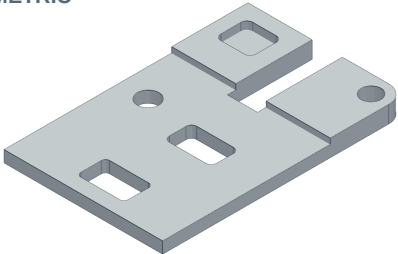
Original CAD

Unmodified input geometry.
Top and isometric views use the same orientation as the meshes below.

TOP DOWN



ISOMETRIC



Views and quality

Top-down meshes are projections of the actual 3D mesh, not separate 2D meshes.
Both densities were independently checked.

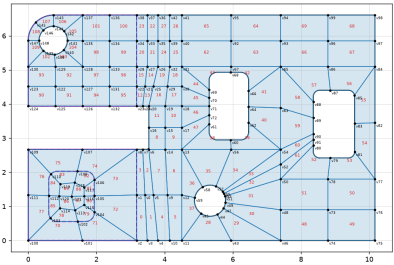
GPT 6 Astra

Counted pass

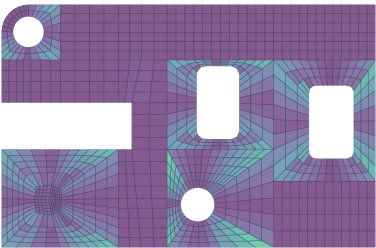
110 blocks | 2,332 hexes
Min SJ: 0.4835
Doubled SJ: 0.4835
1 full revision(s) | 7.15 min
Usage: \$1.97

Standard API equivalent

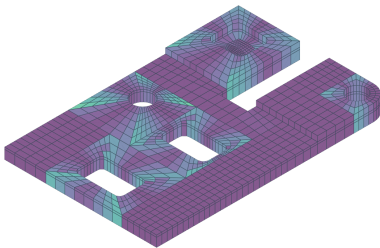
BLOCK LAYOUT



TOP-DOWN MESH

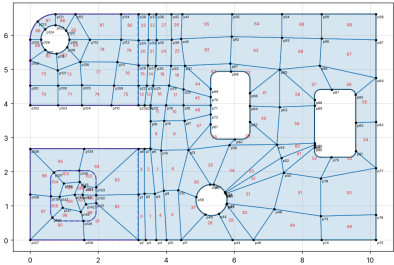


ISOMETRIC MESH



OPTIMISED LAYOUT

Min SJ 0.4835 → 0.6332 | doubled 0.6332



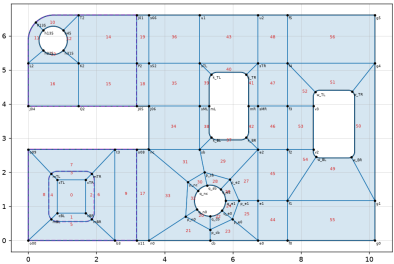
Opus 5.5

Counted pass

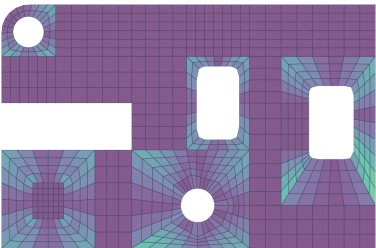
57 blocks | 1,698 hexes
Min SJ: 0.4754
Doubled SJ: 0.4754
1 full revision(s) | 22.60 min
Usage: \$4.53

Claude CLI reported usage

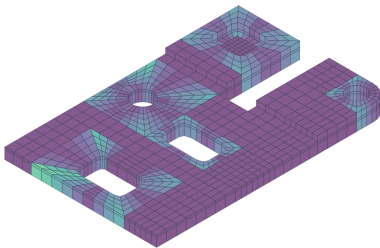
BLOCK LAYOUT



TOP-DOWN MESH

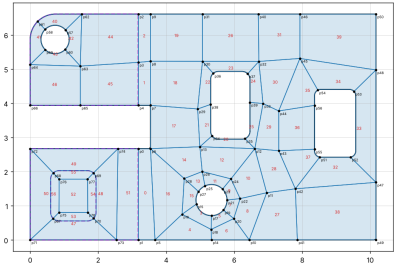


ISOMETRIC MESH



OPTIMISED LAYOUT

Min SJ 0.4754 → 0.6815 | doubled 0.6815



SJ color scale (all meshes)



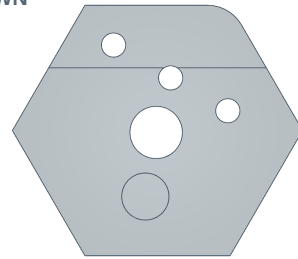
Exposed faces colored by cell SJ; interior cells included in metrics.

Stepped hexagonal body

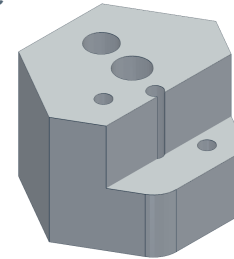
Original CAD

Unmodified input geometry.
Top and isometric views use the same orientation as the meshes below.

TOP DOWN



ISOMETRIC



Views and quality

Top-down meshes are projections of the actual 3D mesh, not separate 2D meshes.
Both densities were independently checked.

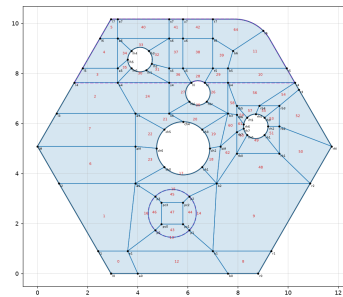
GPT 6 Astra

Counted pass

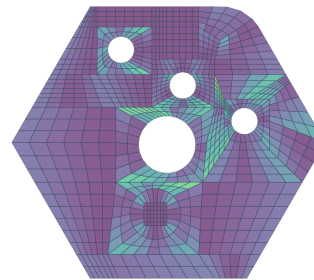
65 blocks | 11,068 hexes
Min SJ: 0.3539
Doubled SJ: 0.3539
1 full revision(s) | 8.02 min
Usage: \$2.04

Standard API equivalent

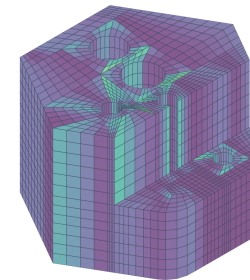
BLOCK LAYOUT



TOP-DOWN MESH

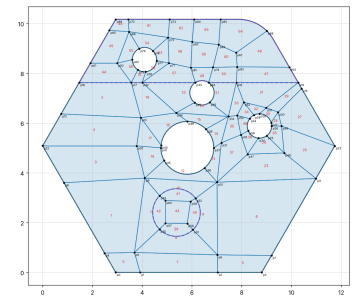


ISOMETRIC MESH



OPTIMISED LAYOUT

Min SJ 0.3539 → 0.5484 | doubled 0.5484



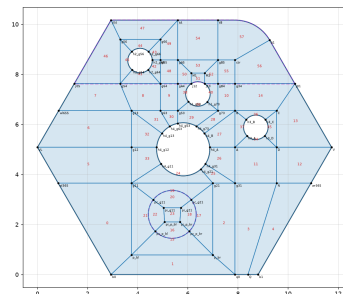
Opus 5.5

Geometry pass / audit rejected

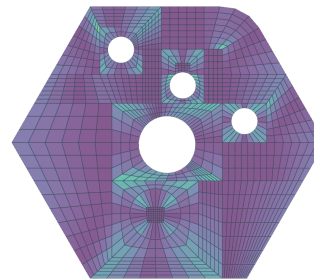
58 blocks | 13,724 hexes
Min SJ: 0.4300
Doubled SJ: 0.4300
2 full revision(s) | 64.97 min
Usage: \$10.12

Claude CLI reported usage

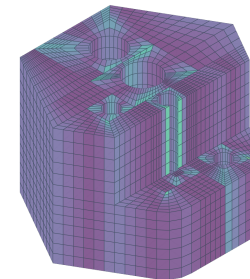
BLOCK LAYOUT



TOP-DOWN MESH

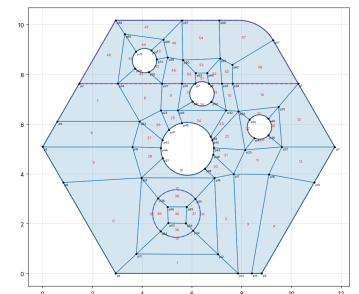


ISOMETRIC MESH



OPTIMISED LAYOUT

Min SJ 0.4300 → 0.5109 | doubled 0.5113



0.2 floor



SJ color scale (all meshes)

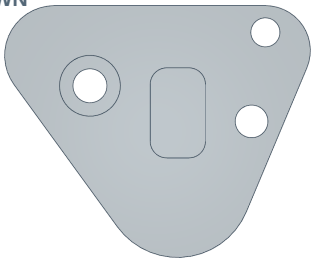
Exposed faces colored by cell SJ; interior cells included in metrics.

Rounded triangular plate

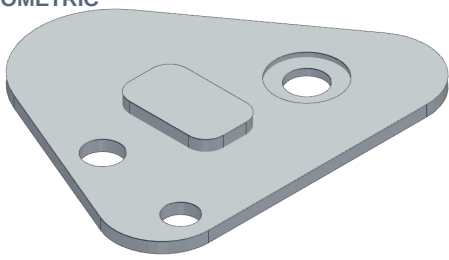
Original CAD

Unmodified input geometry.
Top and isometric views use the same orientation as the meshes below.

TOP DOWN



ISOMETRIC



Views and quality

Top-down meshes are projections of the actual 3D mesh, not separate 2D meshes.
Both densities were independently checked.

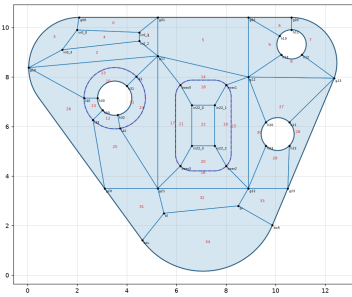
GPT 6 Astra

Counted pass

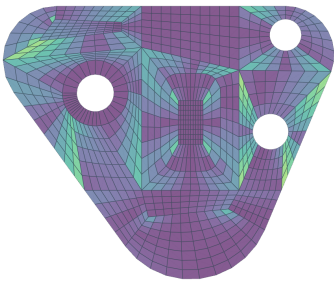
35 blocks | 2,244 hexes
Min SJ: 0.3557
Doubled SJ: 0.3557
1 full revision(s) | 10.86 min
Usage: \$2.59

Standard API equivalent

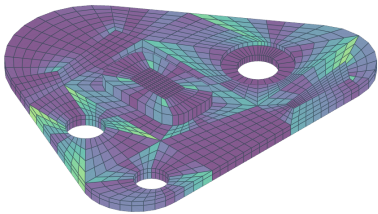
BLOCK LAYOUT



TOP-DOWN MESH

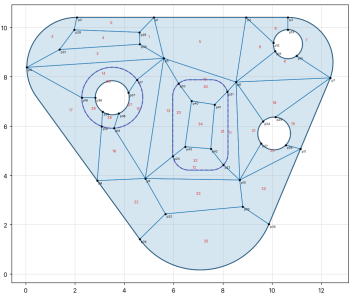


ISOMETRIC MESH



OPTIMISED LAYOUT

Min SJ 0.3557 → 0.4192 | doubled 0.4192



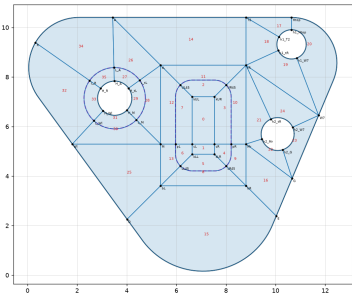
Opus 5.5

Counted pass

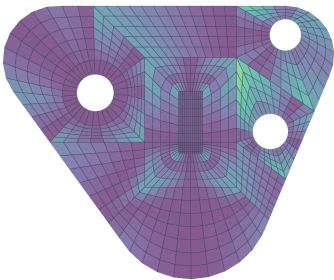
36 blocks | 2,152 hexes
Min SJ: 0.4167
Doubled SJ: 0.4167
1 full revision(s) | 25.53 min
Usage: \$5.45

Claude CLI reported usage

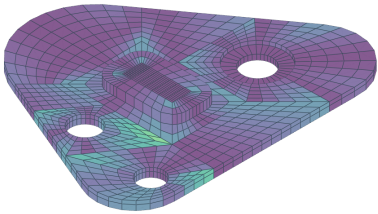
BLOCK LAYOUT



TOP-DOWN MESH

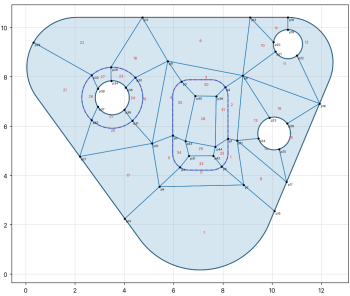


ISOMETRIC MESH



OPTIMISED LAYOUT

Min SJ 0.4167 → 0.5809 | doubled 0.5809



SJ color scale (all meshes)



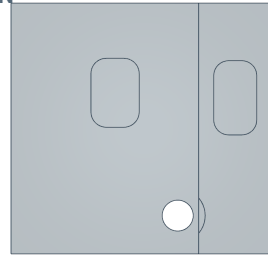
Exposed faces colored by cell SJ; interior cells included in metrics.

Stepped block with pockets

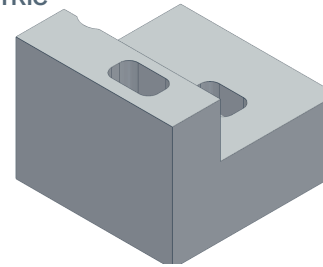
Original CAD

Unmodified input geometry.
Top and isometric views use the same orientation as the meshes below.

TOP DOWN



ISOMETRIC



Views and quality

Top-down meshes are projections of the actual 3D mesh, not separate 2D meshes.
Both densities were independently checked.

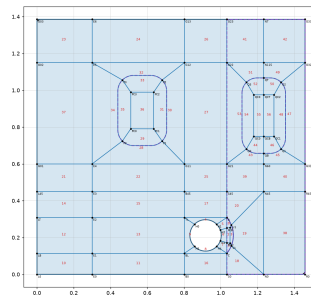
GPT 6 Astra

Counted pass

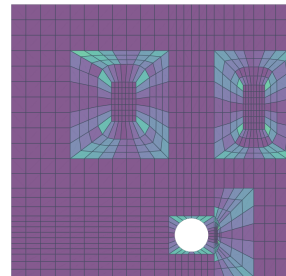
57 blocks | 7,874 hexes
Min SJ: 0.4293
Doubled SJ: 0.4293
3 full revision(s) | 17.96 min
Usage: \$3.81

Standard API equivalent

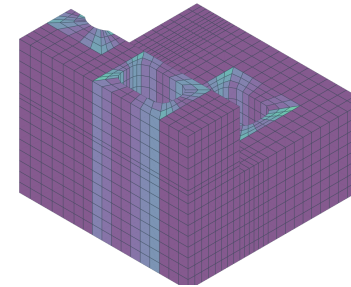
BLOCK LAYOUT



TOP-DOWN MESH

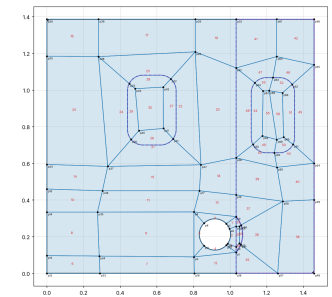


ISOMETRIC MESH



OPTIMISED LAYOUT

Min SJ 0.6092, fails 3-D aspect (21.5 > 20)
Not accepted: the drawn layout stands



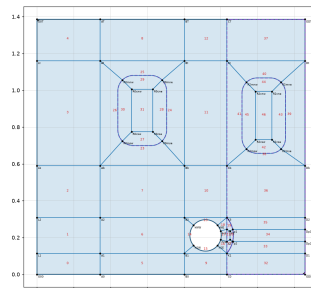
Opus 5.5

Counted pass

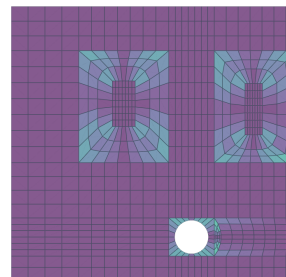
47 blocks | 6,902 hexes
Min SJ: 0.5464
Doubled SJ: 0.5745
1 full revision(s) | 10.94 min
Usage: \$2.34

Claude CLI reported usage

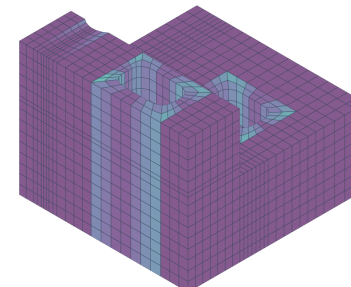
BLOCK LAYOUT



TOP-DOWN MESH

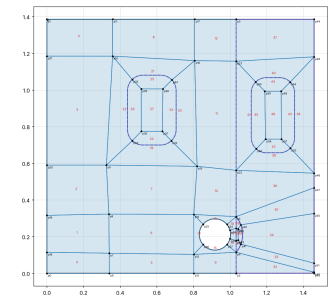


ISOMETRIC MESH



OPTIMISED LAYOUT

Min SJ 0.5464 → 0.5754 | doubled 0.5758



SJ color scale (all meshes)

0.2 floor



Exposed faces colored by cell SJ; interior cells included in metrics.